

Project Report: Avanza - Daily Learning Tracker

1. Executive Summary

Avanza is a full-stack, AI-powered daily learning management system designed to enhance productivity and learning consistency. Built with the MERN stack (React, Node.js, Express, MongoDB), it emphasizes a seamless user experience, intelligent insights, and secure data handling to help users track and achieve their learning objectives.

2. Core Features

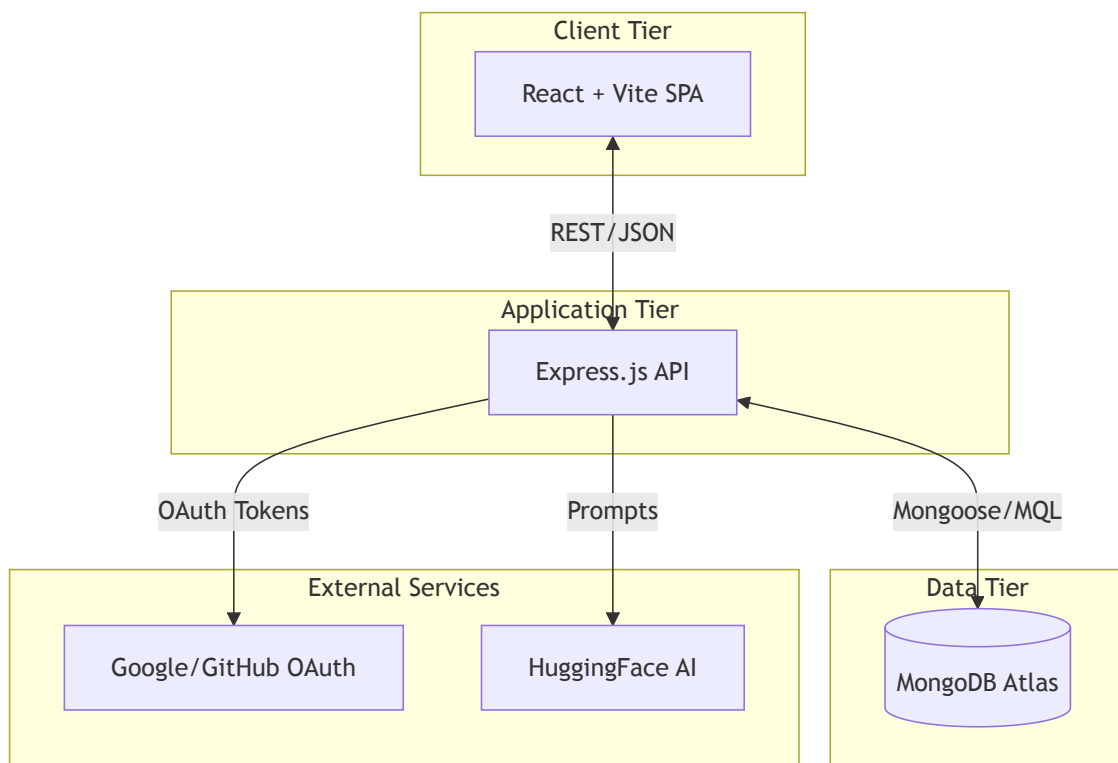
- **Task & Schedule Management:** Users can create, view, and complete tasks with time-spent tracking. Learning objectives are categorized with priorities and color coding.
- **Analytics & Tracking:** Tracks per-objective completion streaks, weekly progress bar charts, and time summaries.
- **AI Integration:** Generates personalized weekly schedules via HuggingFace LLM and features a context-aware AI chat assistant.
- **Notifications:** Real-time reminders and scheduled automated alerts for pending tasks.
- **Security:** Features robust API security (Helmet, NoSQL injection prevention, XSS sanitization) and OAuth/JWT authentication.

3. Technology Stack

Layer	Technologies	Layer	Technologies
Frontend	React 18, Vite, Tailwind CSS, Recharts	AI/External	HuggingFace Inference SDK, Passport.js
Backend	Node.js, Express.js, node-cron	Database	MongoDB + Mongoose ODM
Security	helmet, mongo-sanitize, xss-clean, bcrypt	Auth	JWT, Google/GitHub OAuth

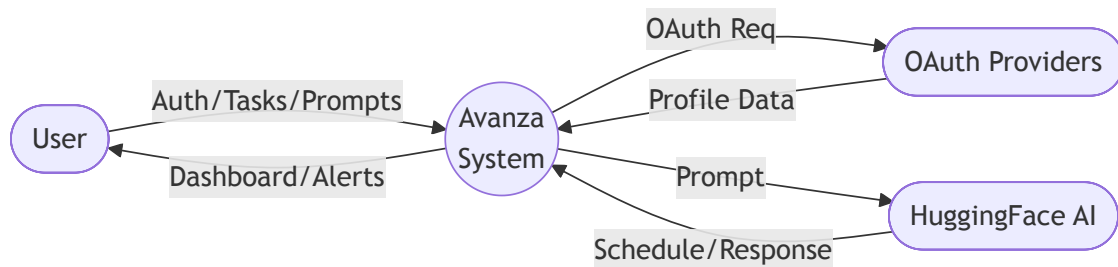
4. System Architecture

The application uses a standard Client-Server architecture utilizing a modern stack, optimized with Vite for high performance.

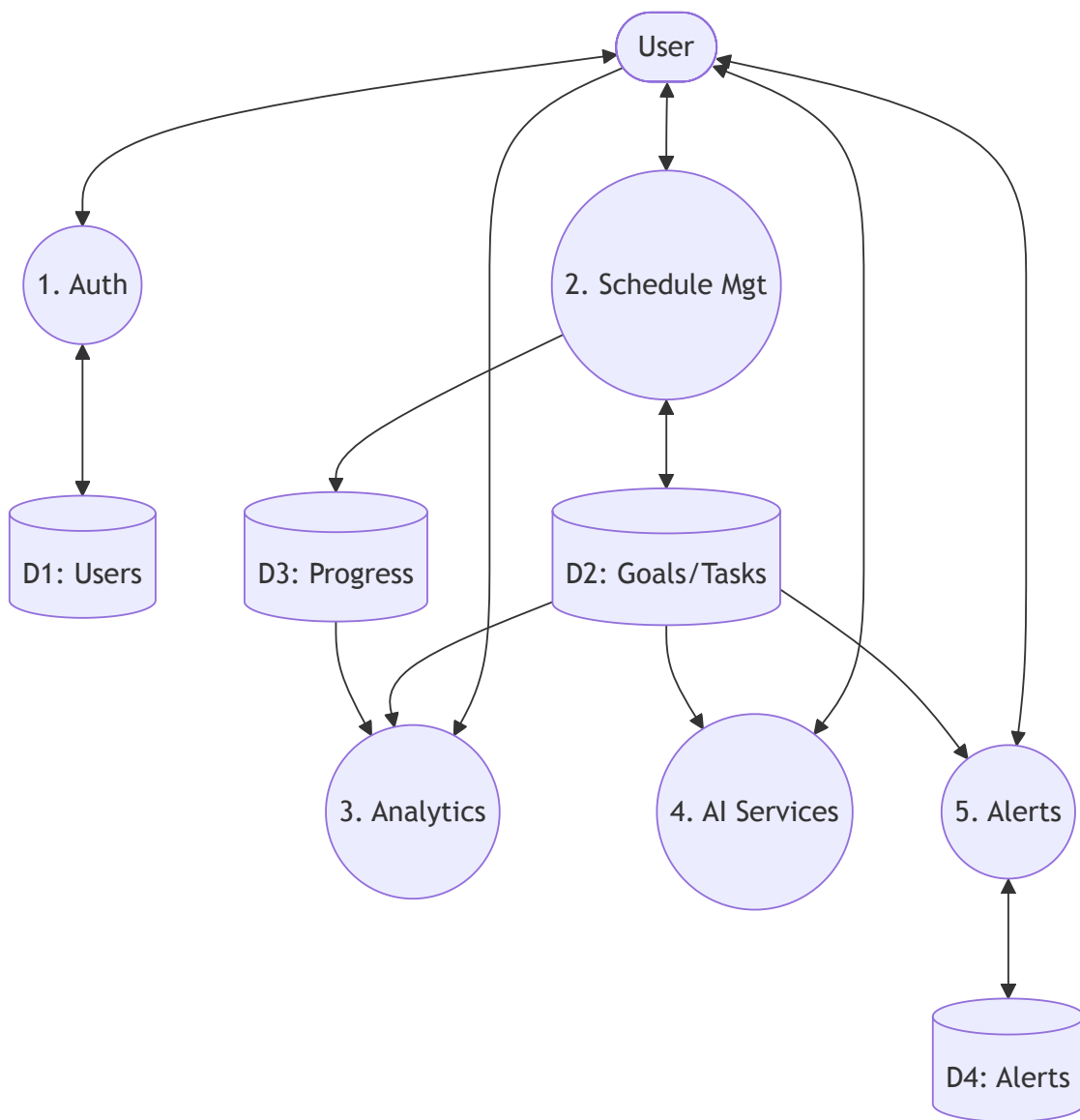


5. Data Flow Diagrams (DFD)

5.1 Context Diagram (Level 0 DFD)



5.2 Logical Data Flow (Level 1 DFD)



6. Database Schema & Security Overview

- **Collections:** Users (profiles/preferences), LearningObjectives (goals), DailyProgress (logs/streaks), Schedules (AI output), Notifications, and Notes .
- **Security:** Hardened API gateway with `express-mongo-sanitize` , `xss-clean` , robust rate limiting (10 req/15 min on auth), and prompt injection guards.